BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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ABSTRACT

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications. BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5%, MultiNLI accuracy to 86.7%, and SQuAD v1.1 question answering Test F1 to 93.2, outperforming human performance by 2.0 points.

1. Introduction & The Unidirectional Limitation

Language model pre-training has shown to be effective for improving many natural language processing tasks. However, standard architectures (such as OpenAI GPT) are unidirectional, where every token can only attend to previous tokens in the self-attention layers. Such restrictions are sub-optimal for sentence-level tasks, and can be devastating when applying fine-tuning based approaches to token-level tasks like SQuAD question answering where it is crucial to incorporate context from both directions.

2. The Masked Language Model (MLM) Paradigm

To train deep bidirectional representations, we propose the Masked Language Model (MLM). In this procedure, we simply mask some percentage of the input tokens at random (15%), and then predict those masked tokens. In addition, we also use a 'next sentence prediction' (NSP) task that jointly pre-trains text-pair representations. Unlike left-to-right language models, the MLM objective enables representations to fuse left and right context.

3. Architecture & Fine-Tuning Recipe

BERT's model architecture is a multi-layer bidirectional Transformer encoder based on the original Transformer. We introduce two model sizes: BERT_BASE (L=12, H=768, A=12, Total Parameters=110M) and BERT_LARGE (L=24, H=1024, A=16, Total Parameters=340M). Fine-tuning BERT is straightforward: for sequence-level classification tasks, we pass the final hidden state of the first token [CLS] into an output classification layer.

4. Experimental Benchmarks on GLUE and SQuAD

BERT achieves state-of-the-art results on 11 NLP tasks. On GLUE, BERT_LARGE attains an 80.5% average score (a 7.7% absolute improvement over existing methods). On SQuAD v1.1, BERT achieves 93.2 Test F1 score, outperforming top ensemble systems.

5. Limitations and Masking Discrepancy

BERT faces notable limitations: First, pre-training creates an artificial discrepancy between pre-training and fine-tuning because the [MASK] token never appears during fine-tuning. Second, BERT assumes masked tokens are conditionally independent given unmasked tokens, failing to model inter-token dependencies among predicted tokens. Third, bi-directional encoding prevents autoregressive token generation for creative generation tasks.

6. Future Horizons

Extensions include autoregressive pre-training (XLNet, RoBERTa removing NSP), span masking (SpanBERT), and parameter reduction through factorized embedding parametrization (ALBERT).